Linux System Administration (E1ITA307)

Linux Infrastructure Deployment Project Report

On

LINUX STORAGE MONITORING SYSTEM

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In partial fulfillment of the award of the degree of

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CERTIFICATE

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1. Introduction

1.1 Aim

The primary aim of this project is to design, develop, and deploy a menu-driven Linux server management simulation representing a Cloud Service Provider on Linux Monitoring Storage System. The system centralizes infrastructure operations—such as multi-tenant client onboarding, Logical Volume Management (LVM) partition provisioning, scheduled archive generation, system health inspection, and remote SSH administration—into a single modular Bash shell management utility.

1.2 Background

Modern cloud infrastructure providers rely on Linux distributions to host tenant environments, allocate dynamic block storage, enforce security boundaries, and monitor system resources. In enterprise systems, maintaining discrete client storage pools and monitoring available storage capacity prevents resource exhaustion and performance degradation. This project simulates core Infrastructure-as-a-Service (IaaS) administrative workflows inside an Ubuntu Server environment using standard Linux utilities, LVM architecture, POSIX permissions, and Bash automation.

1.3 Problem Statement

Small-to-medium infrastructure setups frequently suffer from disjointed administrative workflows: administrators manually partition storage, register client directories without consistent directory permissions, run manual backups, and lack a unified interface for operational checks. Without centralized tooling and automated storage controls, systems risk configuration errors, orphaned storage allocations, unmonitored disk exhaustion, and data loss. A menu-driven Linux infrastructure framework provides a consistent, repeatable interface for client storage monitoring, automation, and operational auditing.

1.4 Objectives

To develop a menu-driven Bash application (cloud_manager.sh) that centralizes system administration tasks.

To implement dynamic block storage management using Logical Volume Management (LVM2) formatted with an ext4 filesystem mounted under /srv/cloud-storage/.

To provision and manage isolated client directories under /srv/cloud-storage/clients/ with strict POSIX permissions.

To automate routine client data backups into compressed .tar.gz archives with validation mechanisms

To configure task scheduling via the cron daemon for hands-off operational maintenance and archival runs.

To monitor system health, block devices, memory, running processes, active SSH access, and operational audit logs.

2. Methodology

2.1 Tools and Technologies Used

2.1 Software Requirements

- Operating System: Ubuntu Server (running on an x86_64 virtualization environment).
- Scripting & Shell: GNU Bash for application logic, looping menus, and administration wrappers**.**
- Volume Management: LVM2 (pvs, vgs, lvs) for physical disk abstraction and dynamic logical volume sizing.
- Filesystem Utilities: ext4, mkfs.ext4, mount, findmnt, df, du.
- Remote Management & Security: OpenSSH Server (sshd), POSIX permissions (chmod, chown, stat).
- Scheduling Daemon: cron (crontab) for recurring automated backups.
- Archival Tools: GNU tar and gzip for backup compression and structural verification.
- Inspection Commands: hostnamectl, uptime, free, ps, top, ss, journalctl.
- **2.2 Hardware Requirements**

- Platform: Virtual machine (e.g., Oracle VirtualBox) configured with Ubuntu Server.
- Primary Virtual Disk **(/dev/sda):** 25 GB allocating the OS root partition (/).

- Secondary Dedicated Storage Disk **(/dev/sdb):** 10 GB virtual block device allocated entirely for physical volume initialization and client storage virtualization.
- Memory & CPU**:** Minimum 2 GB RAM (tested on a 6.2 GB allocated host) and 2 vCPUs.

2.3 Approach / Development Process

- **Step 1:** Scenario Identification & Architecture: Defining the cloud service provider model for multi-tenant storage, package administration, and remote operations.

- **Step 2**: Environment Initialization: Deploying the Ubuntu Server virtual machine, configuring the hostname (cloud-server), and assigning sudo privileges to the administrator account dhruv.

- **Step 3:** Storage Pool Initialization: Initializing /dev/sdb as an LVM Physical Volume (PV), creating the Volume Group cloud-vg, and provisioning a 6 GB Logical Volume client-data formatted as ext4 mounted at /srv/cloud-storage.

- **Step 4:** Security Hardening & Permissions: Creating the client root tree /srv/cloud-storage/clients with directory permission 750 (drwxr-x---).

- **Step 5**: Shell Script Implementation: Developing scripts/cloud_manager.sh using Bash modular functions, input validation, and case...esac menus.

- **Step 6**: Backup Engine & Scheduled Automation: Implementing compressed .tar.gz archive creation with integrity verification and configuring the cron daemon to run backups automatically at 02:00 AM.

- **Step 7**: Diagnostic & Health Instrumentation: Adding diagnostic tools (free -h, df -h, uptime, journalctl -p err) directly into the management interface.

- **Step 8:** Testing and Verification: Running menu validation tests across client listing, storage checks, archive inspection, package verification, and security boundaries.

3. Linux System Administration – Design and Implementation

3.1 System Architecture

1. The system architecture illustrates the overall design of the Cloud Service Provider project, which simulates Linux server management using a menu-driven Bash application. The administrator interacts with the system through a terminal and selects different operations from cloud_manager.sh.
2. The application provides modules for client management, storage and LVM, backup and recovery, package management, server monitoring, SSH security, automation, and system logs. These modules interact with Linux services running on an Ubuntu Server environment.
3. The storage layer uses Logical Volume Management (LVM) and the ext4 file system to manage client data. Backup scripts create compressed .tar.gz archives, while Cron supports scheduled automation. Monitoring commands collect system information, resource usage, and log details.
4. Overall, the architecture demonstrates how different Linux administration tasks can be integrated into a single menu-driven application, making server management easier to understand and perform in a local virtual-machine environment.

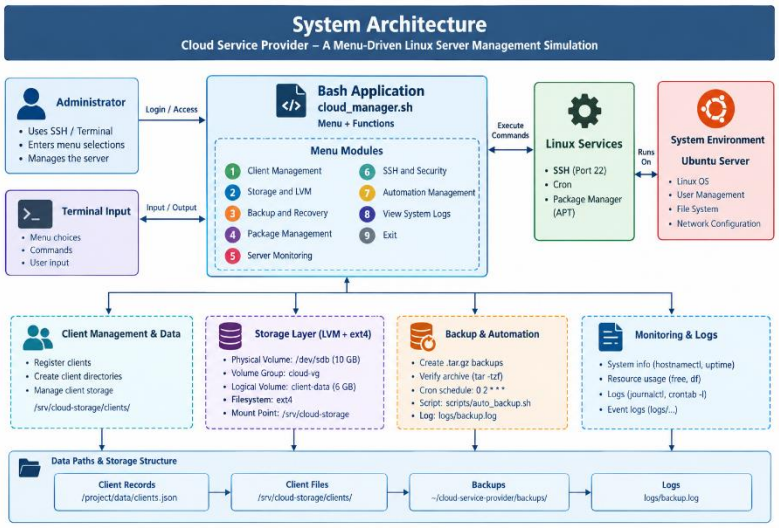


Figure 3.1: System Architecture of Linux Storage Monitoring System

Components and their roles

Components and Their Roles

1. Administrator: Interacts with the system through SSH or the terminal and selects the required server management operations.

2. Terminal Interface: Accepts user input, displays the menu options, and shows command outputs.

3. Bash Application (`cloud_manager.sh`): Acts as the core component that connects the menu options with the corresponding Linux administration tasks.

4. Client Management Module: Registers clients, creates client directories, and manages client-related data.

5. Storage and LVM Module: Manages logical storage volumes using LVM and the ext4 file system to provide storage for client data.

6. Backup and Recovery Module: Creates compressed `.tar.gz` backup archives and supports archive verification and recovery operations.

7. Package Management Module: Uses the APT package manager to install and manage required Linux software packages.

8. Server Monitoring Module: Displays system information, uptime, memory usage, and disk space using Linux commands.

9. SSH and Security Module: Supports remote server access through SSH and helps manage file permissions and access.

10. Automation Management Module: Uses Cron jobs and shell scripts to schedule repetitive tasks, such as automatic backups.

11. Monitoring and Logging Module: Uses system logs and application log files to track events, errors, and backup activities.

12. Ubuntu Server Environment: Provides the operating system and underlying Linux services required to execute the application and its management tasks.

3.2 Linux Environment Setup

The operational environment was provisioned on an Ubuntu Linux virtual server inside Oracle VirtualBox. System identity, kernel architecture, and administrative privileges were verified using core diagnostic tools

```
whoami
hostnamectl
```

```
dhruv
 Static hostname: cloud-server
       Icon name: computer-vm
         Chassis: vm
      Machine ID: c0cc61162ce648f58e8b268ae7337537
  Virtualization: oracle
Operating System: Ubuntu Linux
          Kernel: Linux 6.8.0-generic
    Architecture: x86-64
```

The application directory structure is maintained under /home/dhruv/cloud-service-provider/, organizing scripts, data files, backup archives, and logs into clear functional paths

```
/home/dhruv/cloud-service-provider/
├── backups/
├── data/
├── logs/
└── scripts/
```

3.3 User and Group Management

Multi-tenant security requires strict boundary management between system administrators and client workloads:

- Administrative execution is handled via the account dhruv, configured with administrative sudo privileges.

- The central client directory structure restricts unprivileged access by enforcing POSIX user/group boundaries.
- Client directories inherit access permissions that restrict traversal across tenant boundaries, ensuring tenant isolation within the virtual machine.

3.4 File and Directory Management

Tenant files are decoupled from the root OS volume and mounted to /srv/cloud-storage. Inside this mount point, a dedicated client directory tree isolates each customer's data:

```
sudo mkdir -p /srv/cloud-storage/clients
sudo chmod 750 /srv/cloud-storage/clients
```

When new tenants are registered, dedicated folders are provisioned under /srv/cloud-storage/clients/<client_name>/. Directory verification via du -sh confirms baseline tenant footprint initialization

```
du -sh /srv/cloud-storage/clients/client01
```

```
4.0K    /srv/cloud-storage/clients/client01
```

3.5 Package and Process Management

The package management module verifies whether the required Linux packages are installed. These packages provide SSH remote access, scheduled task management, LVM volume orchestration, and archive creation:

- openssh-server: Remote terminal management daemon.
- cron: Task scheduling daemon.
- lvm2: User-space tools for Logical Volume Management.
- tar: Archive creation and gzip compression utility.

```
dpkg -s openssh-server cron lvm2 tar | grep -E "Package|Status"
```

The execution output confirms all four required packages are installed and active:

```
========== PACKAGE MANAGEMENT ==========
Package status:
openssh-server: installed
cron: installed
lvm2: installed
tar: installed

Install missing packages with:
sudo apt install openssh-server cron lvm2 tar

Press Enter to continue...
```

Figure 3.2: Package Management Module verification confirming installation of openssh-server, cron, lvm2, and tar.

3.6 Storage and File System Configuration

The project implements a 3-tier LVM architecture on a secondary virtual drive (/dev/sdb, ~10 GB) to support flexible resizing and storage separation:

1. Physical Volume (PV): Initialized directly on /dev/sdb.
2. Volume Group (VG): Named cloud-vg, aggregating the storage pool.
3. Logical Volume (LV): Named client-data, provisioned at 6 GB.
4. Filesystem: Formatted as ext4 and mounted persistently to /srv/cloud-storage.

3.6.1 Storage Management Module Execution

The interactive management script encapsulates storage operations under menu option 2 (Storage and LVM). Within this module, the administrator can query filesystem capacity, calculate individual tenant storage usage, and inspect active LVM volumes without manually typing complex administrative commands.

```
Bash

========== STORAGE MANAGEMENT ==========
1. Show filesystem capacity
2. Show client storage usage
3. Show LVM configuration
4. Return
```

Option 1: Filesystem Capacity Check

Queries the target mount /srv/cloud-storage using df -h:

```
Filesystem                        Size  Used Avail Use% Mounted on
/dev/mapper/cloud--vg-client--data  5.9G  1.6M  5.6G   1% /srv/cloud-storage
```

Figure 3.3: Storage Management Submenu – Filesystem capacity check for /srv/cloud-storage.

Option 2: Individual Client Storage Consumption

Prompts for the client identifier (e.g., client01) and computes disk usage using du -sh:

```
Choose [1-4]: 2
Client name: client01
4.0K    /srv/cloud-storage/clients/client01
```

Figure 3.4: Storage Management Submenu – Client directory storage utilization inspection.

Option 3: LVM Architecture Inspection

Executes privileged volume management commands (pvs, vgs, lvs) to report storage pool allocations:

```
  Choose [1-4]: 3
  PV         VG        Fmt  Attr PSize   PFree
  /dev/sda3  ubuntu-vg lvm2 a--  <38.00g 19.00g
  /dev/sdb   cloud-vg  lvm2 a--  <10.00g <4.00g
  VG        #PV #LV #SN Attr   VSize   VFree
  cloud-vg    1   1   0 wz--n- <10.00g <4.00g
  ubuntu-vg   1   1   0 wz--n- <38.00g 19.00g
  LV          VG        Attr       LSize   Pool Origin Data% Meta% Move Log Cpy%Sy
  client-data cloud-vg  -wi-ao----   6.00g
  ubuntu-lv   ubuntu-vg -wi-ao---- <19.00g
```

Figure 3.5: Storage Management Submenu – LVM physical volume, volume group, and logical volume allocations

3.6.2 Terminal Verification of LVM and Mount Points

Administrative verification commands were executed directly from the terminal shell (lsblk, pvs, vgs, lvs, df -h, and findmnt) to confirm kernel-level device mappings:

```
  └─ubuntu--vg-ubuntu--lv 252:0    0    19G  0 lvm  /
sdb                         8:16   0    10G  0 disk
└─cloud--vg-client--data  252:1    0     6G  0 lvm  /srv/cloud-storage

dhruv@cloud-server:~/Desktop$ sudo pvs
  PV         VG        Fmt  Attr PSize   PFree
  /dev/sda3  ubuntu-vg lvm2 a--  <38.00g 19.00g
  /dev/sdb   cloud-vg  lvm2 a--  <10.00g <4.00g

dhruv@cloud-server:~/Desktop$ sudo vgs
  VG        #PV #LV #SN Attr   VSize   VFree
  cloud-vg    1   1   0 wz--n- <10.00g <4.00g
  ubuntu-vg   1   1   0 wz--n- <38.00g 19.00g

dhruv@cloud-server:~/Desktop$ sudo lvs
  LV          VG        Attr       LSize   Pool Origin Data% Meta% Move Log Cpy%Sync
  client-data cloud-vg  -wi-ao----   6.00g
  ubuntu-lv   ubuntu-vg -wi-ao---- <19.00g

dhruv@cloud-server:~/Desktop$ df -h /srv/cloud-storage
Filesystem                         Size  Used Avail Use% Mounted on
/dev/mapper/cloud--vg-client--data  5.9G  1.6M  5.6G   1% /srv/cloud-storage

dhruv@cloud-server:~/Desktop$ findmnt /srv/cloud-storage
TARGET             SOURCE                                FSTYPE OPTIONS
/srv/cloud-storage /dev/mapper/cloud--vg-client--data ext4   rw,relatime
```

Figure 3.6: Command-line verification of storage block devices (lsblk), LVM allocations (pvs, vgs, lvs), and persistent mount points (findmnt, df)

3.7 Backup and Automation

3.7.1 Backup Strategy & Verification

Backups are created per client using GNU tar combined with gzip compression. To protect against data corruption, archives are verified immediately after creation via the -tzf flag before completing the operation:

```
tar -czf ~/cloud-service-provider/backups/client01_20261021_043946.tar.gz \
    -C /srv/cloud-storage/clients client01
tar -tzf ~/cloud-service-provider/backups/client01_20261021_043946.tar.gz
```

The menu interface shows the creation and integrity verification of the backup archive:

```
========== BACKUP MANAGEMENT ==========
Client name to back up: client01
Backup created: /home/dhruv/cloud-service-provider/backups/client01_20261021_043946.
Archive integrity: OK

Press Enter to continue...
```

Figure 3.7: Backup Management Module – Creation and integrity verification of tenant archive client01_20261021_043946.tar.gz.

3.7.2 Scheduled Automation (Cron)

Cron automates recurring administrative tasks. A daily job was configured to run the backup script, verify the archive, and record the result in a log:

```
crontab -l
```

The automation management module confirms that the cron daemon is active and displays the configured user crontab job:

```
========== AUTOMATION MANAGEMENT ==========
--- Cron service ---
active

--- Current user's scheduled jobs ---
0 2 * * * /home/dhruv/cloud-service-provider/scripts/auto_backup.sh

Edit scheduled jobs with: crontab -e

Press Enter to continue...
```

Figure 3.8: Automation Management Module – Active cron service status and daily 02:00 AM backup schedule.

3.8 Security and Access Control

OpenSSH provides the remote administration service. The client directory uses restrictive permissions so that users outside its owner and group have no directory access.

The SSH and Security module performs automated checks across current credentials, remote listening sockets, and directory security permissions:

```
========== SSH AND SECURITY ==========
--- Current user ---
dhruv

--- SSH service ---
active

--- SSH listening sockets ---
LISTEN 0    4096    0.0.0.0:22    0.0.0.0:*    users:(("sshd",pid=1395,fd=3),("syste
LISTEN 0    4096       [::]:22       [::]:*    users:(("sshd",pid=1395,fd=4),("syste

--- Client directory permissions ---
drwxr-x--- 3 dhruv dhruv 4096 Oct 21 03:48 /srv/cloud-storage/clients

Review users with: cut -d: -f1 /etc/passwd
Review firewall with: sudo ufw status

Press Enter to continue...
```

Figure 3.9: SSH and Security Module – Daemon listener on port 22 and restricted 750 mode on /srv/cloud-storage/clients

4. Results and Testing

4.1 Implementation Results

4.1.1 Client Listing and Verification

The client management module allows the system administrator to query the central registry and inspect all active registered tenants hosted in the cloud storage system. By selecting Option 2 (List clients) from the interactive CLIENT MANAGEMENT submenu, the script reads existing customer records stored in the data repository and verifies the presence of active client accounts:

```
========== CLIENT MANAGEMENT ==========
1. Register client
2. List clients
3. Return
Choose [1-3]: 2
client01

Press Enter to continue...
```

Figure 4.1: Client Management Submenu – Listing existing registered tenants (client01).

4.1.2 Server Health and Resource Monitoring

The server monitoring module provides an overview of server health, including memory, CPU info, disk usage, and recent system errors. Administrators can use this information to identify potential issues before they cause service downtime:

```
========== SERVER MONITORING ==========
--- Hostname and uptime ---
cloud-server
 04:45:21 up 47 min,  1 user,  load average: 0.08, 0.02, 0.01

--- Memory ---
               total        used        free      shared  buff/cache   available
Mem:           2.6Gi       808Mi       1.1Gi       8.1Mi       866Mi       1.8Gi
Swap:          2.9Gi          0B       2.9Gi

--- Filesystem ---
Filesystem                         Size  Used Avail Use% Mounted on
tmpfs                              530M  1.3M  529M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv   19G  7.7G   10G  44% /
tmpfs                              1.3G     0  1.3G   0% /dev/shm
none                               1.0M     0  1.0M   0% /run/credentials/systemd-j
tmpfs                              1.3G  8.0K  1.3G   1% /tmp
none                               1.0M     0  1.0M   0% /run/credentials/systemd-r
/dev/sda2                          2.0G  192M  1.6G  11% /boot
/dev/mapper/cloud--vg-client--data  5.9G  1.6M  5.6G   1% /srv/cloud-storage
none                               1.0M     0  1.0M   0% /run/credentials/systemd-n
none                               1.0M     0  1.0M   0% /run/credentials/getty@tty
tmpfs                              265M   60K  265M   1% /run/user/1000

--- CPU information ---
2
```

```
--- Recent system errors ---
[sudo: authenticate] Password:
Oct 21 03:56:52 cloud-server kernel: KHO: Failed to reserve lowmem scratch buffer
Oct 21 03:57:13 cloud-server kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems
Oct 21 03:57:13 cloud-server kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configu
Oct 21 03:57:13 cloud-server kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switc
Oct 21 03:57:48 cloud-server lightdm[1678]: gkr-pam: unable to locate daemon control

Press Enter to continue...
```

Figure 4.2: Server Monitoring Module – System load, memory allocation, filesystem metrics, and kernel diagnostic logs.

4.1.3 Operational System Logs

System logs provide a timestamped record of important operations. They help administrators review client registration and backup activity to maintain an audit trail.

```
========== SYSTEM LOG ==========
2026-10-21 03:48:34 | Client registered: client01
2026-10-21 03:49:11 | Backup created for client01
2026-10-21 04:02:27 | Backup created for client01
2026-10-21 04:03:40 | Backup created for client01
2026-10-21 04:39:46 | Backup created for client01

Press Enter to continue...
```

Figure 4.3: System Logs Module – Timestamped execution logs capturing client registration and backup runs.

4.2 Testing and Validation

Cloud Storage Management System
Test Cases and Results

Test Case	Module / Action	Expected Result	Actual Result	Status
TC-01	**Client Listing** Option 1 / 2	Displays all active registered clients from clients.txt.	Client client01 was listed and verified.	Pass
TC-02	**Storage Filesystem Check** Option 2 / 1	Displays the capacity and usage of /srv/cloud-storage.	Reported 5.9 GB total, 1.6 MB used, 5.6 GB available, and 1% usage.	Pass
TC-03	**Client Directory Usage** Option 2 / 2	Calculates the selected client's directory size using du -sh.	Client client01 storage footprint was evaluated as 4.0 KB.	Pass
TC-04	**LVM Architecture Validation** Option 2 / 3	Displays physical volumes, volume groups, and logical volumes.	cloud-vg and the 6.00 GB client-data logical volume were displayed.	Pass
TC-05	**Backup Creation and Testing** Option 3	Creates a .tar.gz archive and verifies its integrity using tar -tzf.	Backup archive was created successfully; integrity check returned OK.	Pass
TC-06	**Package Dependency Audit** Option 4	Checks the installation status of SSH, cron, LVM2, and tar.	All four required packages were verified as installed.	Pass
TC-07	**System Health Inspection** Option 5	Displays uptime, memory usage, filesystem information, and error logs.	System diagnostics were displayed correctly without script errors.	Pass
TC-08	**Security and Remote Access** Option 6	Verifies SSH service status, port 22, and directory permissions.	SSH was active, port 22 was listening, and permissions were verified as drwxr-x---.	Pass
TC-09	**Automated Task Inspection** Option 7	Checks the cron service and daily backup schedule at 2:00 AM.	Cron was active, and the 0 2 * * * ... auto_backup.sh entry was confirmed.	Pass
TC-10	**Audit Log Review** Option 8	Displays recent operations with valid timestamps.	Client registration and backup operations appeared in the audit logs.	Pass

Overall Test Summary

Total Test Cases	Passed	Failed
10	10	0

Testing result: All 10 test cases passed according to the supplied results, giving a 100% pass rate.

Note: The results above are based on the test data provided. They have not been independently executed or verified.

5. Conclusion and Future Scope

5.1 Conclusion

To conclude, our Cloud Service Provider project demonstrates core Linux server administration using Ubuntu Server and Bash shell scripting. We implemented client management, dedicated LVM storage, package checks, compressed backups, Cron automation, SSH status checks, server monitoring, and system logging.

We tested the main operations and verified backup archive integrity, storage configuration, service status, and client permissions. This is a local Linux server simulation that demonstrates the fundamental administration tasks involved in managing cloud infrastructure.

5.2 Future Scope

- LVM Dynamic Extension: Add menu options to execute online volume expansion via lvextend and resize2fs when storage pools fill up.
- Disaster Recovery Automation: Implement automated backup restoration workflows (tar -xzf) that include checksum verification (e.g., sha256sum).
- Firewall Hardening: Integrate dynamic ufw configuration menus to limit SSH access to trusted IP subnets.
- Web Dashboard: Build a browser-based monitoring dashboard using Flask to display live storage, CPU, and memory metrics.
- Multi-Drive RAID Pools: Transition physical volume provisioning from single virtual disks to software RAID arrays (e.g., mdadm RAID 1/5) for hardware fault tolerance.

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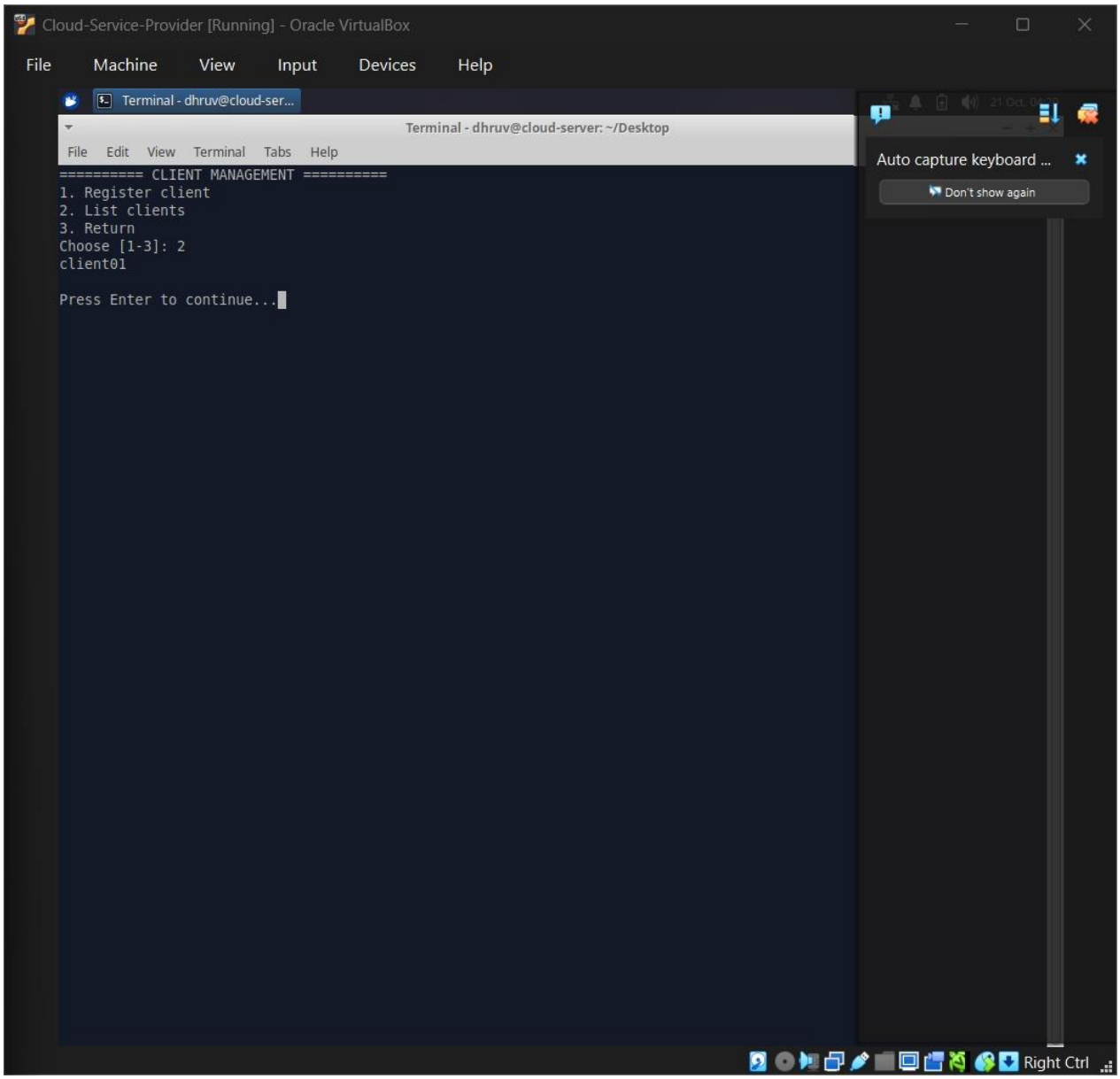
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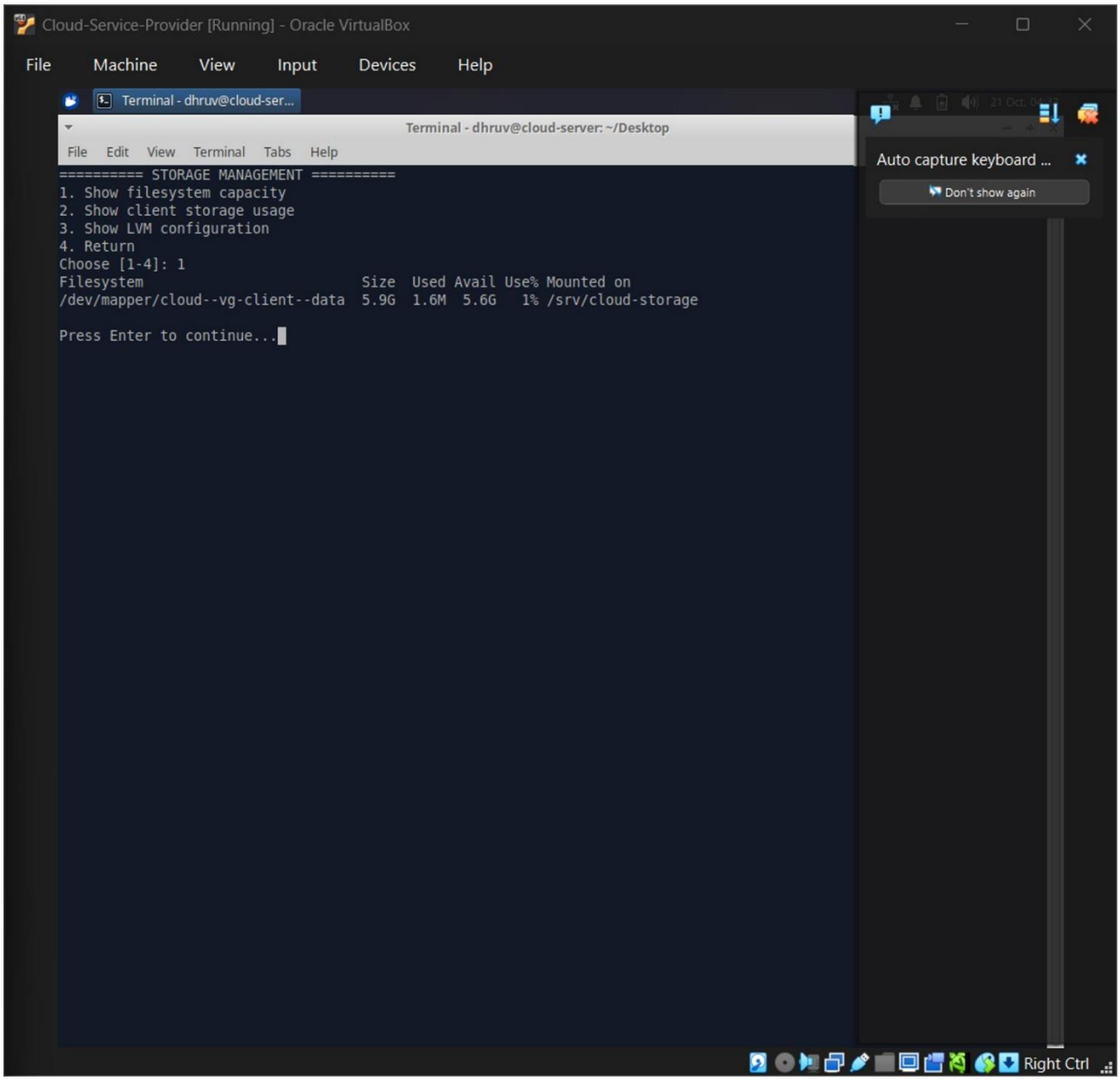
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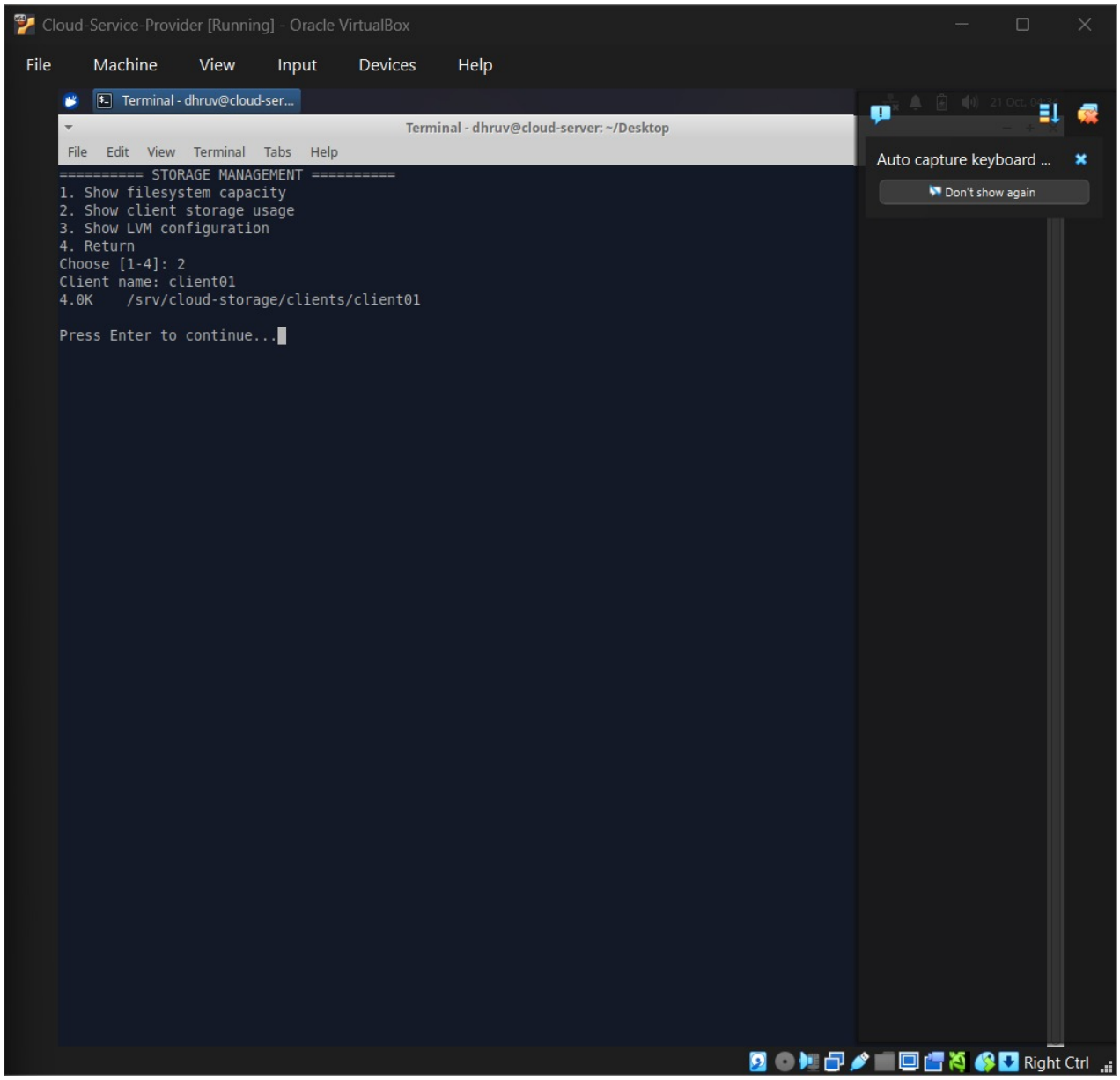
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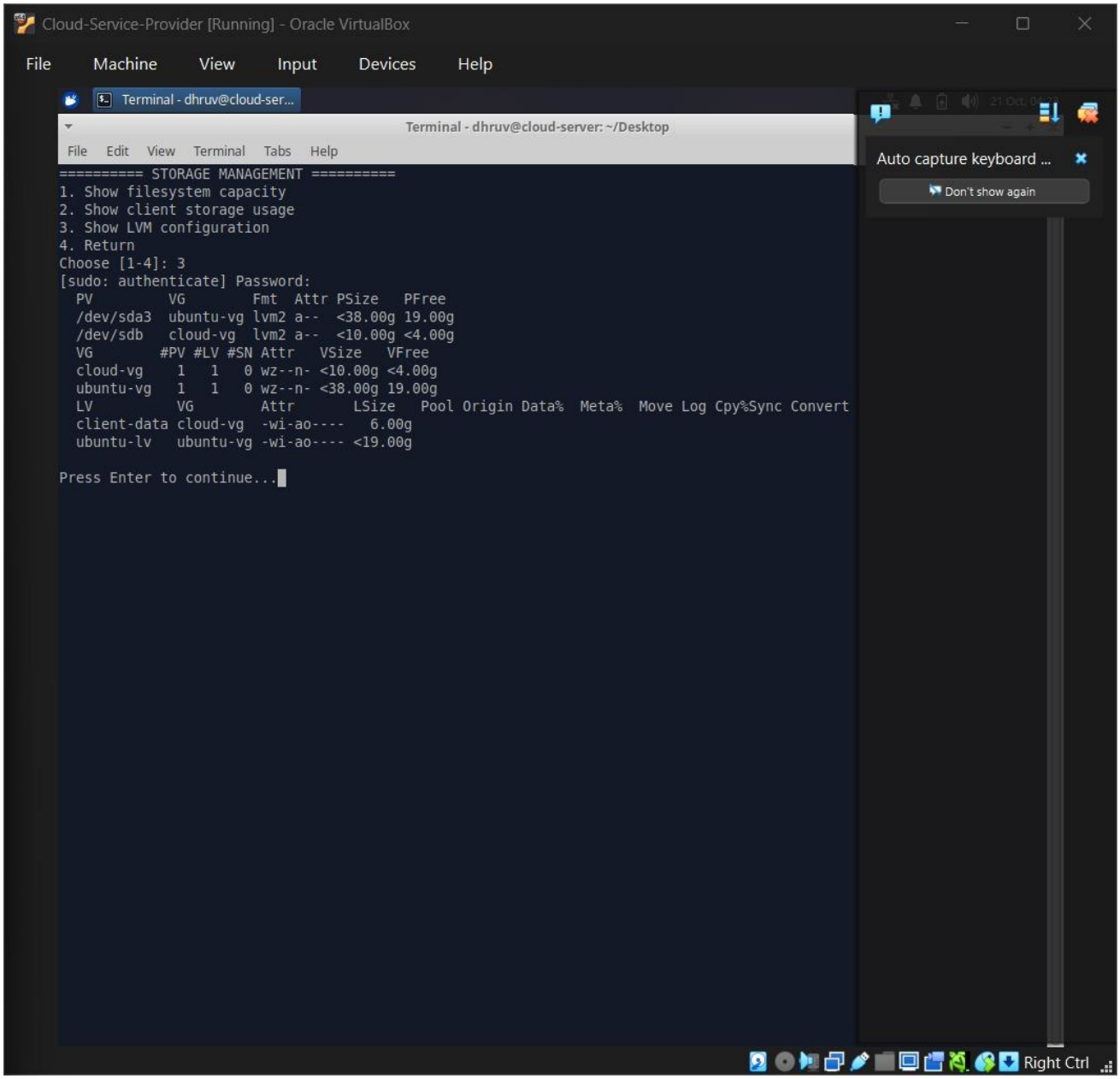
Appendices

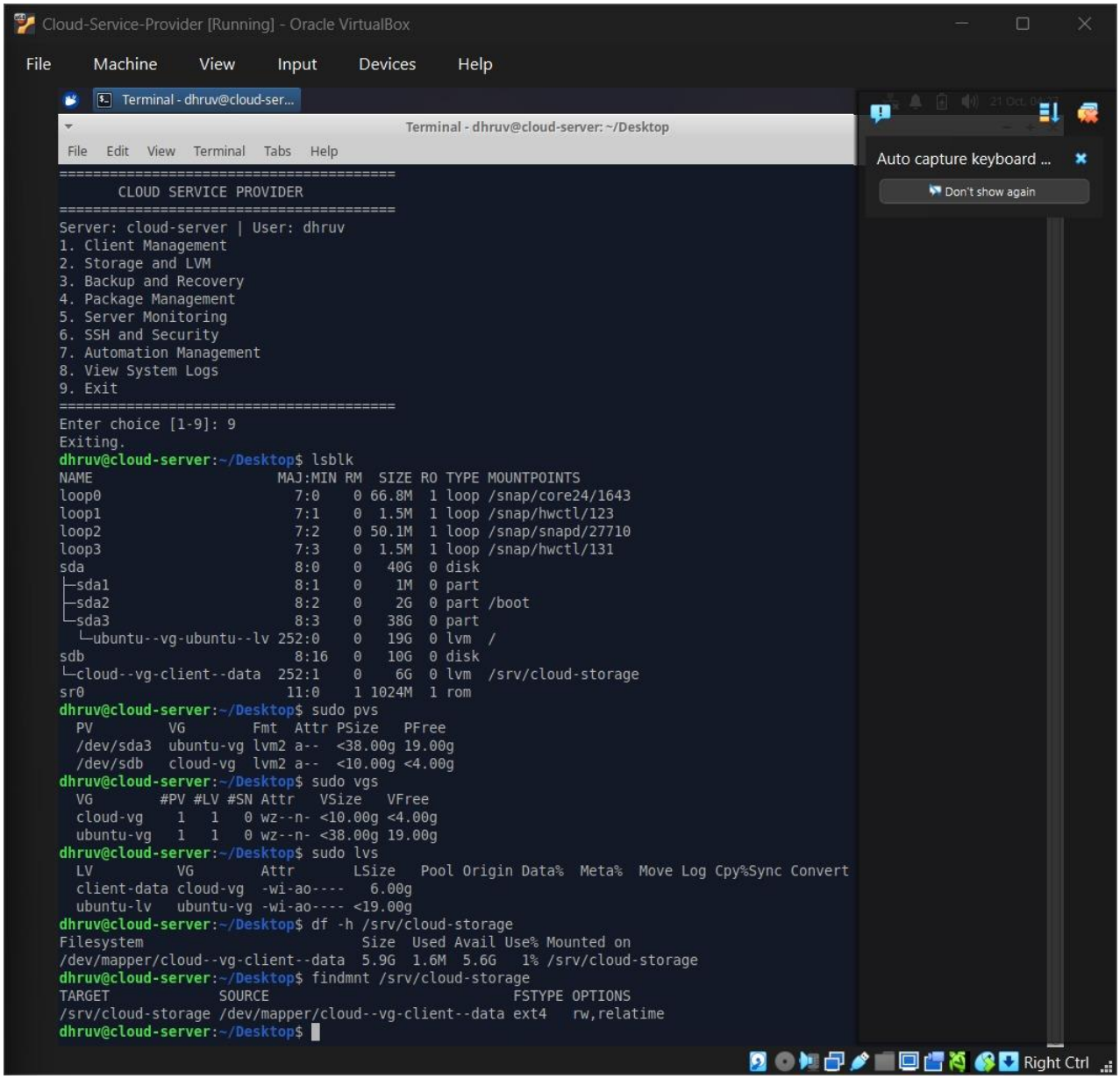
Appendix A: Screenshot of Project Output :

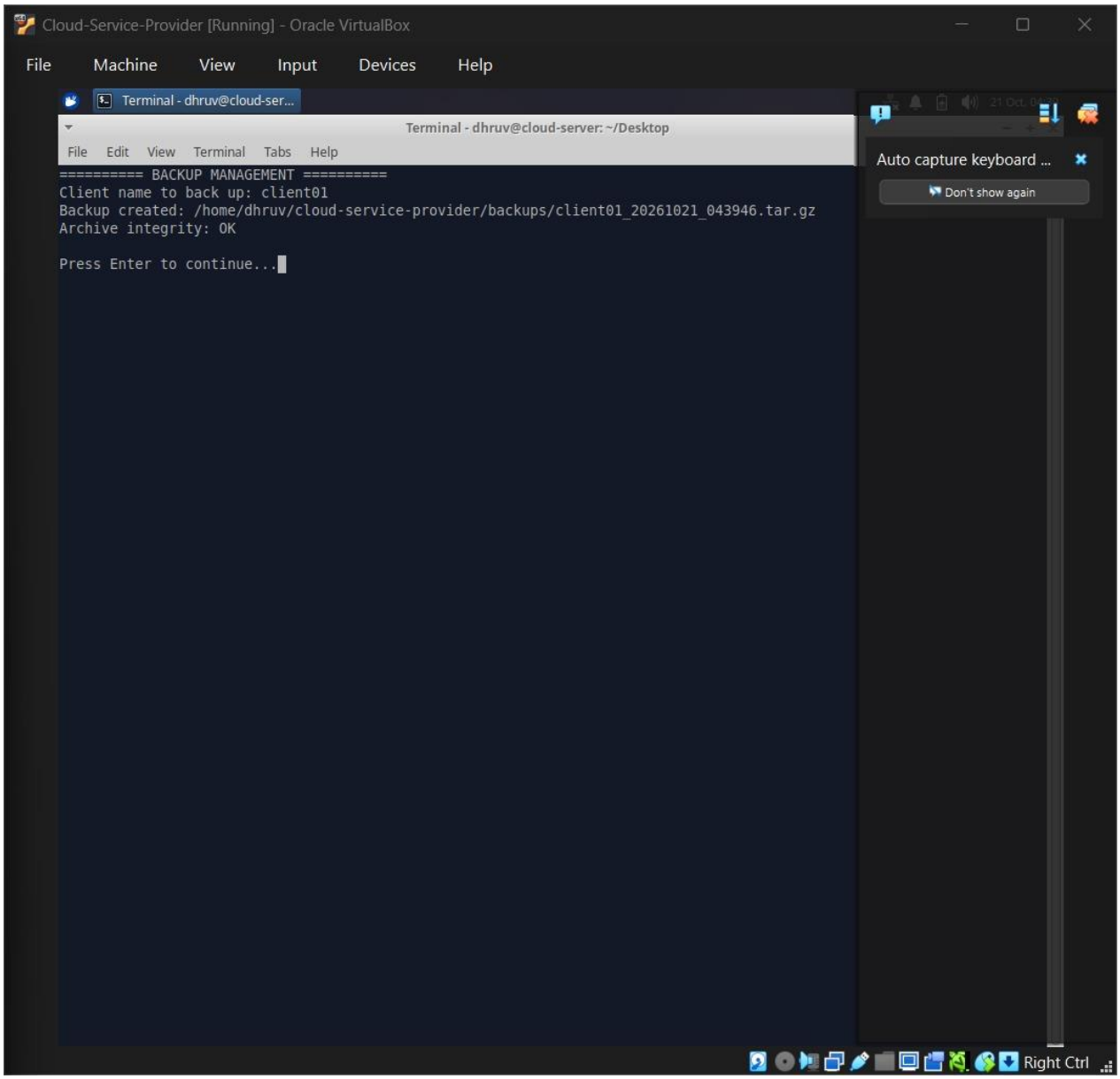


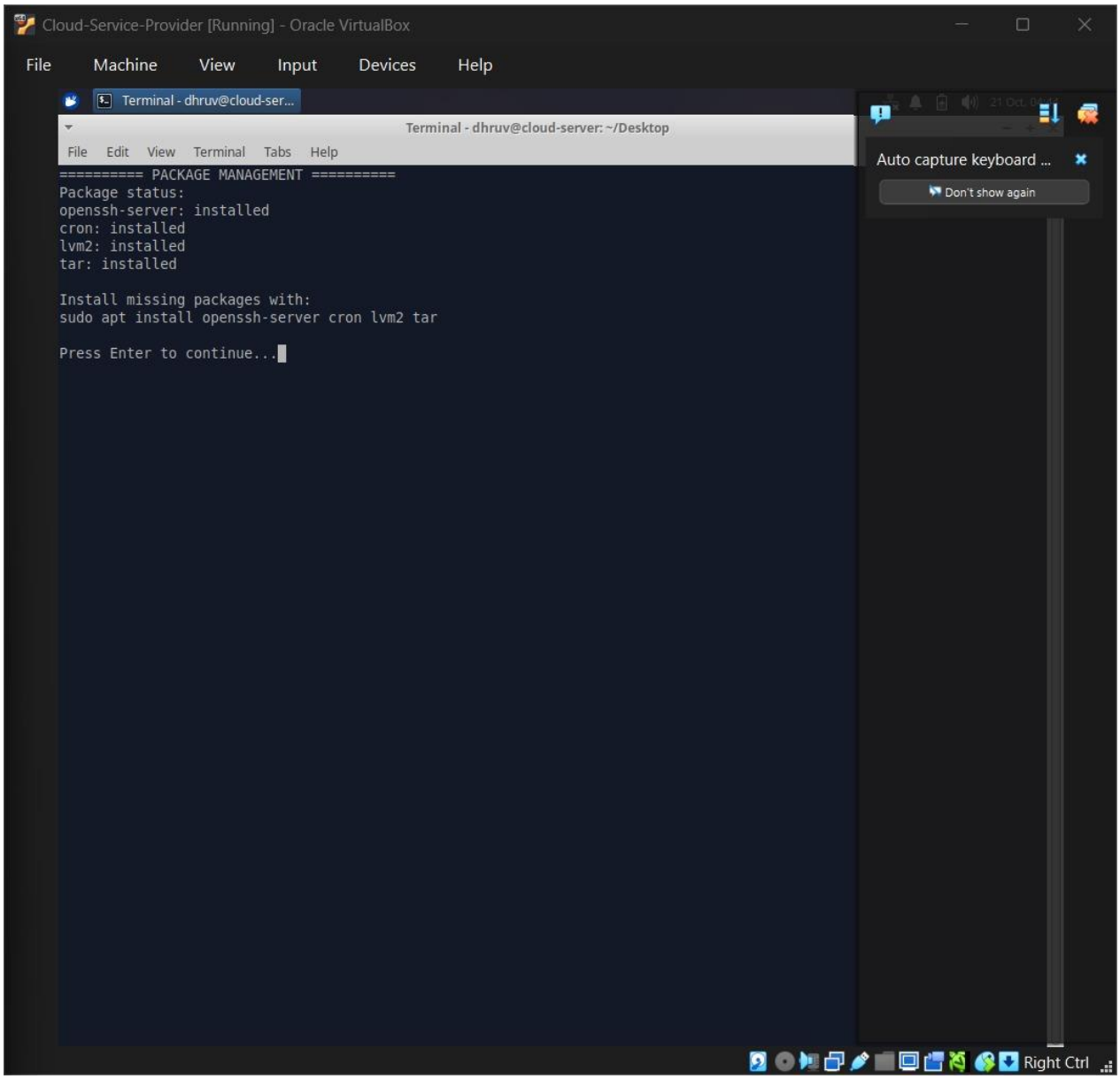

Cloud-Service-Provider [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Terminal - dhruv@cloud-ser...
Terminal - dhruv@cloud-server: ~/Desktop
File Edit View Terminal Tabs Help
========== STORAGE MANAGEMENT ==========
1. Show filesystem capacity
2. Show client storage usage
3. Show LVM configuration
4. Return
Choose [1-4]: 1
Filesystem Size Used Avail Use% Mounted on
/dev/mapper/cloud--vg-client--data 5.9G 1.6M 5.6G 1% /srv/cloud-storage
Press Enter to continue...
Auto capture keyboard ...
Don't show again
Right Ctrl

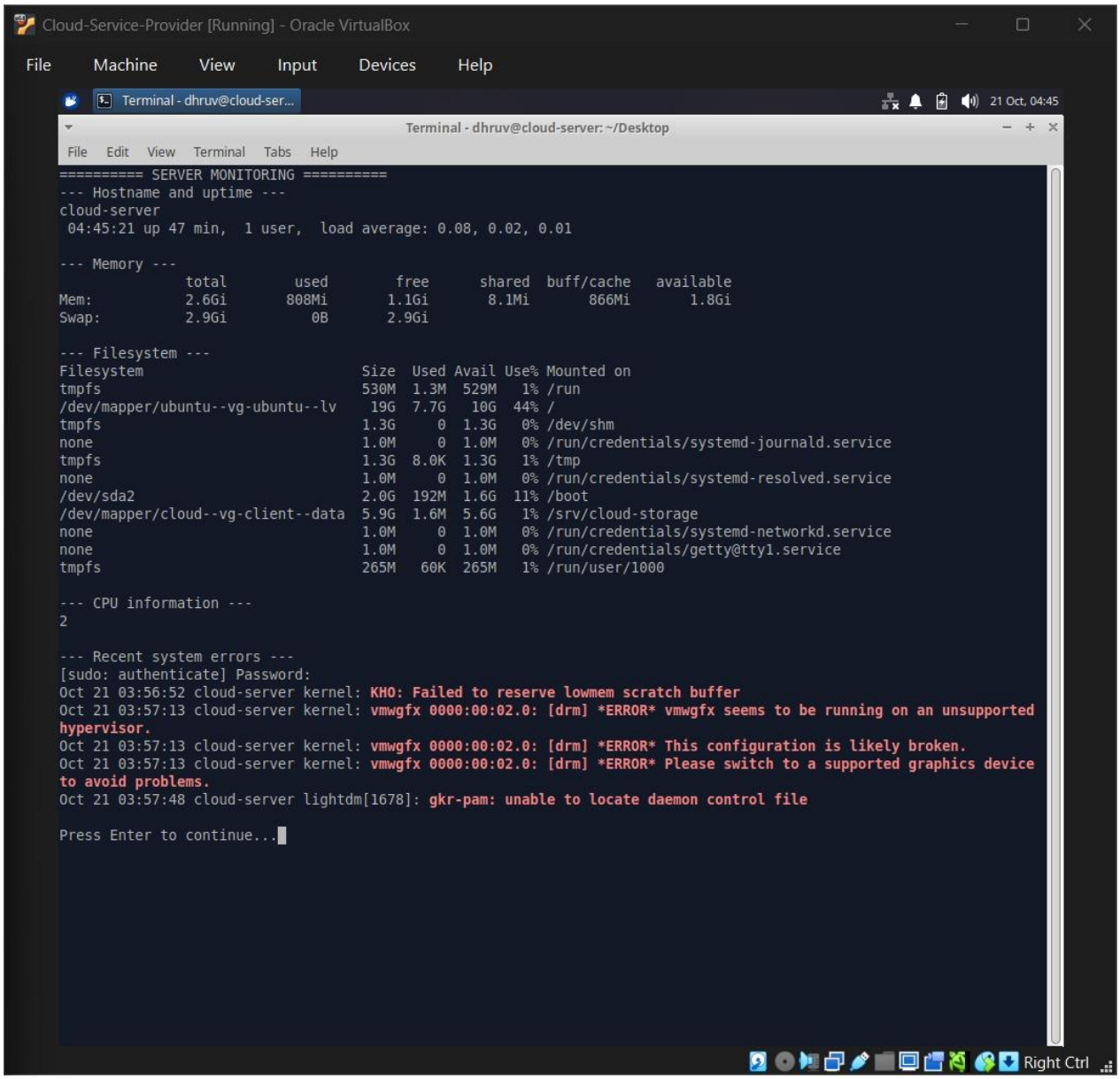

Cloud-Service-Provider [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Terminal - dhruv@cloud-ser...
Terminal - dhruv@cloud-server: ~/Desktop
File Edit View Terminal Tabs Help
========== STORAGE MANAGEMENT ==========
1. Show filesystem capacity
2. Show client storage usage
3. Show LVM configuration
4. Return
Choose [1-4]: 2
Client name: client01
4.0K /srv/cloud-storage/clients/client01
Press Enter to continue...
Auto capture keyboard ...
Don't show again
Right Ctrl

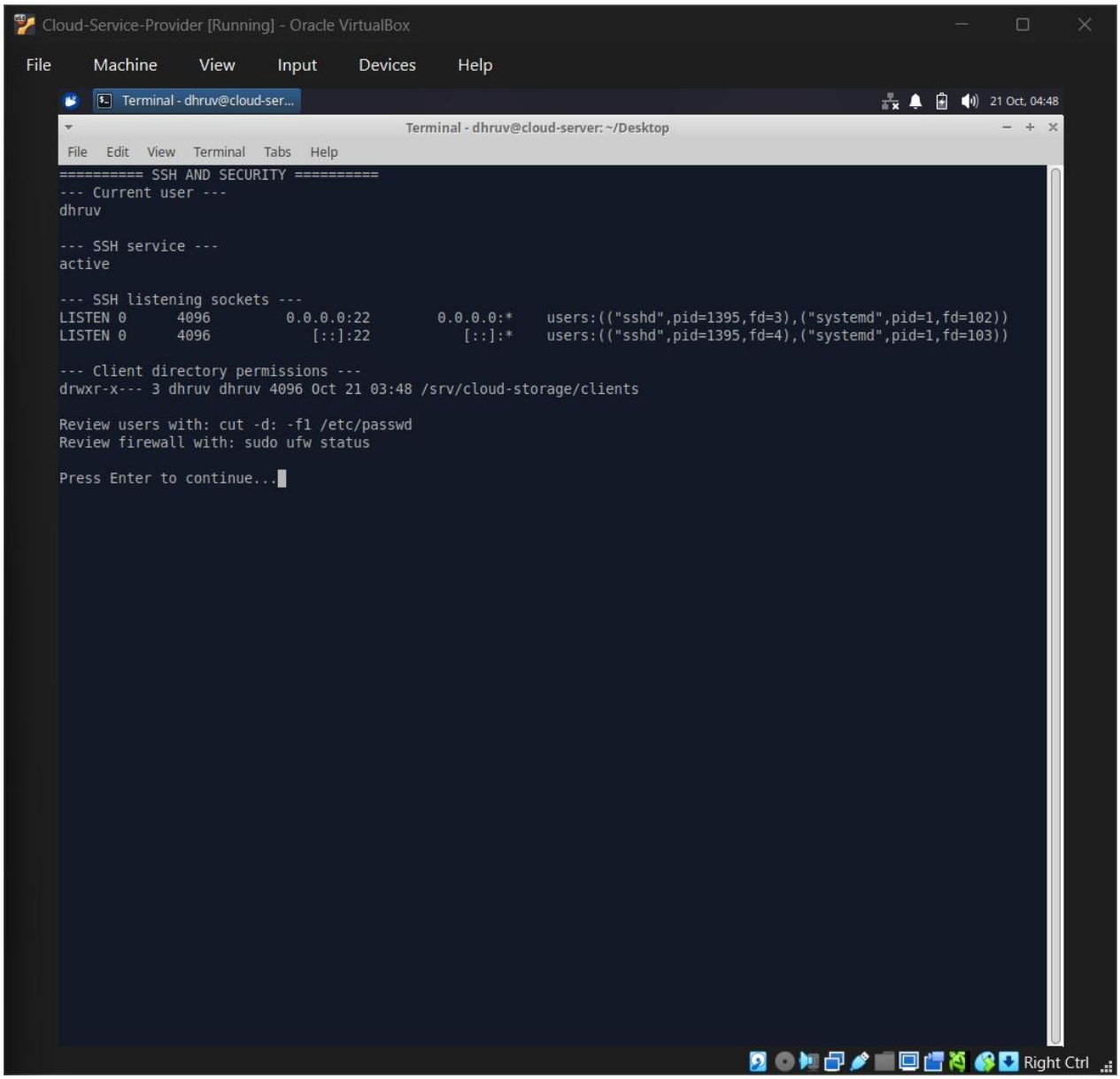

Cloud-Service-Provider [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Terminal - dhruv@cloud-ser...
Terminal - dhruv@cloud-server: ~/Desktop
File Edit View Terminal Tabs Help
========== STORAGE MANAGEMENT ==========
1. Show filesystem capacity
2. Show client storage usage
3. Show LVM configuration
4. Return
Choose [1-4]: 3
[sudo: authenticate] Password:
PV VG Fmt Attr PSize PFree
/dev/sda3 ubuntu-vg lvm2 a-- <38.00g 19.00g
/dev/sdb cloud-vg lvm2 a-- <10.00g <4.00g
VG #PV #LV #SN Attr VSize VFree
cloud-vg 1 1 0 wz--n- <10.00g <4.00g
ubuntu-vg 1 1 0 wz--n- <38.00g 19.00g
LV VG Attr LSize Pool Origin Data% Meta% Move Log Cpy%Sync Convert
client-data cloud-vg -wi-ao---- 6.00g
ubuntu-lv ubuntu-vg -wi-ao---- <19.00g
Press Enter to continue...
Auto capture keyboard ...
Don't show again
Right Ctrl

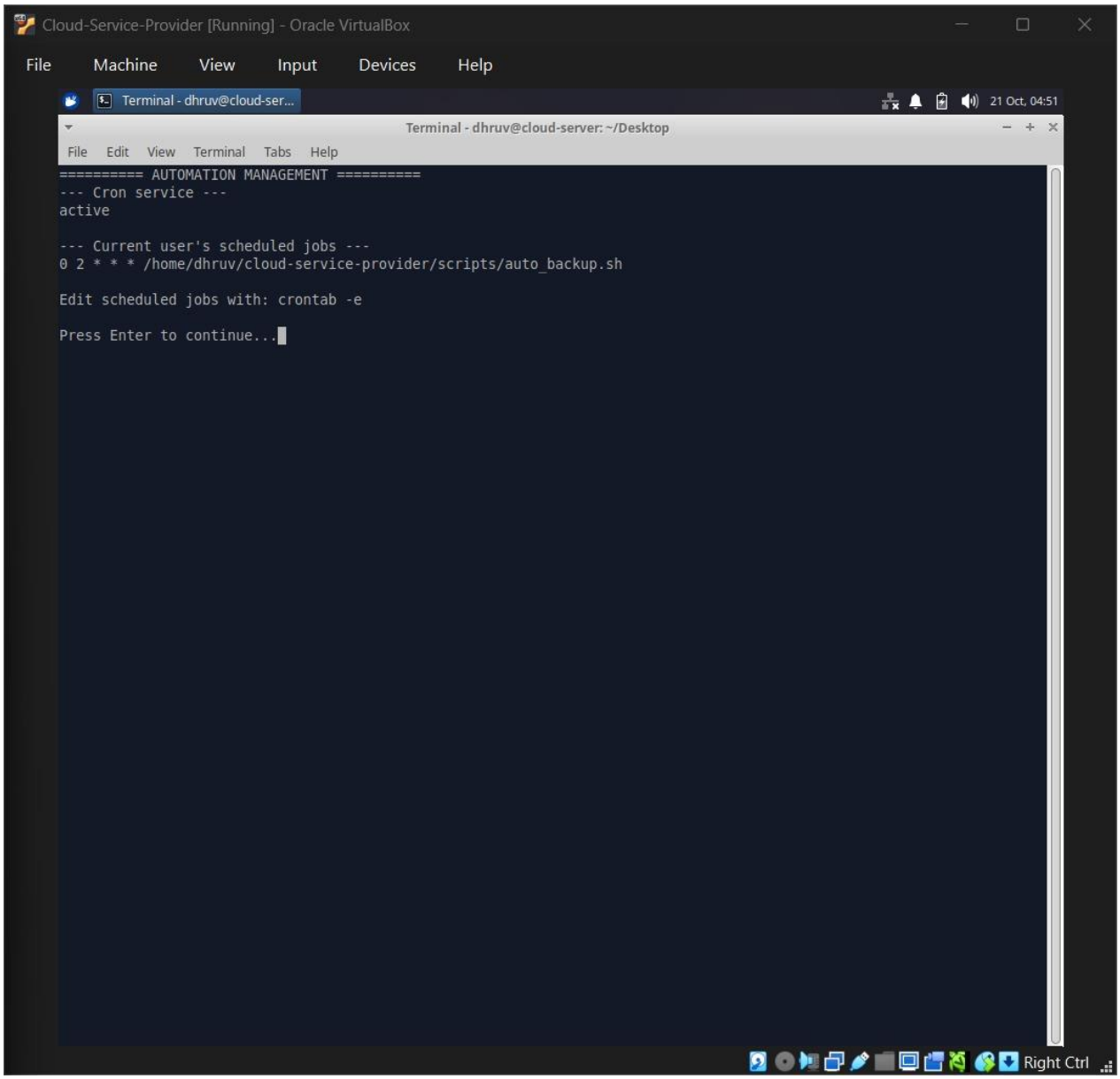

Cloud-Service-Provider [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Terminal - dhruv@cloud-ser...
Terminal - dhruv@cloud-server: ~/Desktop
File Edit View Terminal Tabs Help
CLOUD SERVICE PROVIDER
Server: cloud-server | User: dhruv
1. Client Management
2. Storage and LVM
3. Backup and Recovery
4. Package Management
5. Server Monitoring
6. SSH and Security
7. Automation Management
8. View System Logs
9. Exit
Enter choice [1-9]: 9
Exiting.
dhruv@cloud-server:~/Desktop$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0 7:0 0 66.8M 1 loop /snap/core24/1643
loop1 7:1 0 1.5M 1 loop /snap/hwctl/123
loop2 7:2 0 50.1M 1 loop /snap/snapd/27710
loop3 7:3 0 1.5M 1 loop /snap/hwctl/131
sda 8:0 0 40G 0 disk
├─sda1 8:1 0 1M 0 part
├─sda2 8:2 0 2G 0 part /boot
└─sda3 8:3 0 38G 0 part
└─ubuntu--vg-ubuntu--lv 252:0 0 19G 0 lvm /
sdb 8:16 0 10G 0 disk
└─cloud--vg-client--data 252:1 0 6G 0 lvm /srv/cloud-storage
sr0 11:0 1 1024M 1 rom
dhruv@cloud-server:~/Desktop$ sudo pvs
PV VG Fmt Attr PSize PFree
/dev/sda3 ubuntu-vg lvm2 a-- <38.00g 19.00g
/dev/sdb cloud-vg lvm2 a-- <10.00g <4.00g
dhruv@cloud-server:~/Desktop$ sudo vgs
VG #PV #LV #SN Attr VSize VFree
cloud-vg 1 1 0 wz--n- <10.00g <4.00g
ubuntu-vg 1 1 0 wz--n- <38.00g 19.00g
dhruv@cloud-server:~/Desktop$ sudo lvs
LV VG Attr LSize Pool Origin Data% Meta% Move Log Cpy%Sync Convert
client-data cloud-vg -wi-ao---- 6.00g
ubuntu-lv ubuntu-vg -wi-ao---- <19.00g
dhruv@cloud-server:~/Desktop$ df -h /srv/cloud-storage
Filesystem Size Used Avail Use% Mounted on
/dev/mapper/cloud--vg-client--data 5.9G 1.6M 5.6G 1% /srv/cloud-storage
dhruv@cloud-server:~/Desktop$ findmnt /srv/cloud-storage
TARGET SOURCE FSTYPE OPTIONS
/srv/cloud-storage /dev/mapper/cloud--vg-client--data ext4 rw,relatime
dhruv@cloud-server:~/Desktop$
Auto capture keyboard ...
Don't show again
Right Ctrl

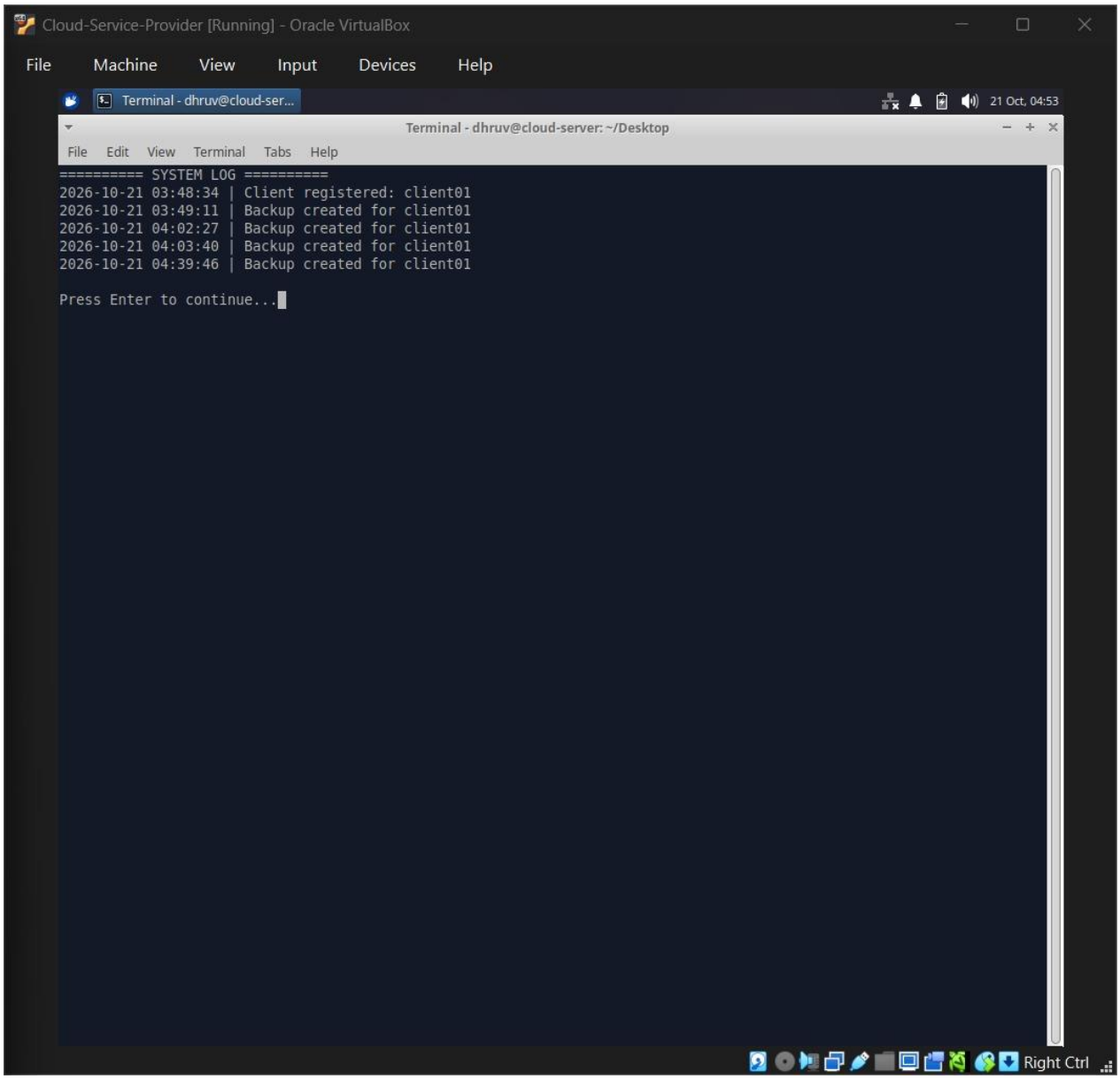

Cloud-Service-Provider [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Terminal - dhruv@cloud-ser...
Terminal - dhruv@cloud-server: ~/Desktop
File Edit View Terminal Tabs Help
========== BACKUP MANAGEMENT ==========
Client name to back up: client01
Backup created: /home/dhruv/cloud-service-provider/backups/client01_20261021_043946.tar.gz
Archive integrity: OK
Press Enter to continue...
Auto capture keyboard ...
Don't show again
Right Ctrl


Cloud-Service-Provider [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Terminal - dhruv@cloud-ser...
Terminal - dhruv@cloud-server: ~/Desktop
File Edit View Terminal Tabs Help
========== PACKAGE MANAGEMENT ==========
Package status:
openssh-server: installed
cron: installed
lvm2: installed
tar: installed
Install missing packages with:
sudo apt install openssh-server cron lvm2 tar
Press Enter to continue...
Auto capture keyboard ...
Don't show again
Right Ctrl


Cloud-Service-Provider [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Terminal - dhruv@cloud-ser...
21 Oct, 04:45
Terminal - dhruv@cloud-server: ~/Desktop
File Edit View Terminal Tabs Help
========== SERVER MONITORING ==========
--- Hostname and uptime ---
cloud-server
04:45:21 up 47 min, 1 user, load average: 0.08, 0.02, 0.01
--- Memory ---
total used free shared buff/cache available
Mem: 2.6Gi 808Mi 1.1Gi 8.1Mi 866Mi 1.8Gi
Swap: 2.9Gi 0B 2.9Gi
--- Filesystem ---
Filesystem Size Used Avail Use% Mounted on
tmpfs 530M 1.3M 529M 1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 19G 7.7G 10G 44% /
tmpfs 1.3G 0 1.3G 0% /dev/shm
none 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
tmpfs 1.3G 8.0K 1.3G 1% /tmp
none 1.0M 0 1.0M 0% /run/credentials/systemd-resolved.service
/dev/sda2 2.0G 192M 1.6G 11% /boot
/dev/mapper/cloud--vg-client--data 5.9G 1.6M 5.6G 1% /srv/cloud-storage
none 1.0M 0 1.0M 0% /run/credentials/systemd-networkd.service
none 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service
tmpfs 265M 60K 265M 1% /run/user/1000
--- CPU information ---
2
--- Recent system errors ---
[sudo: authenticate] Password:
Oct 21 03:56:52 cloud-server kernel: KHO: Failed to reserve lowmem scratch buffer
Oct 21 03:57:13 cloud-server kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.
Oct 21 03:57:13 cloud-server kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
Oct 21 03:57:13 cloud-server kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.
Oct 21 03:57:48 cloud-server lightdm[1678]: gkr-pam: unable to locate daemon control file
Press Enter to continue...
Right Ctrl


Cloud-Service-Provider [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Terminal - dhruv@cloud-ser...
21 Oct, 04:48
Terminal - dhruv@cloud-server: ~/Desktop
File Edit View Terminal Tabs Help
========== SSH AND SECURITY ==========
--- Current user ---
dhruv
--- SSH service ---
active
--- SSH listening sockets ---
LISTEN 0 4096 0.0.0.0:22 0.0.0.0:* users:(("sshd",pid=1395,fd=3),("systemd",pid=1,fd=102))
LISTEN 0 4096 [::]:22 [::]:* users:(("sshd",pid=1395,fd=4),("systemd",pid=1,fd=103))
--- Client directory permissions ---
drwxr-x--- 3 dhruv dhruv 4096 Oct 21 03:48 /srv/cloud-storage/clients
Review users with: cut -d: -f1 /etc/passwd
Review firewall with: sudo ufw status
Press Enter to continue...
Right Ctrl


Cloud-Service-Provider [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Terminal - dhruv@cloud-ser...
21 Oct, 04:51
Terminal - dhruv@cloud-server: ~/Desktop
File Edit View Terminal Tabs Help
========== AUTOMATION MANAGEMENT ==========
--- Cron service ---
active
--- Current user's scheduled jobs ---
0 2 * * * /home/dhruv/cloud-service-provider/scripts/auto_backup.sh
Edit scheduled jobs with: crontab -e
Press Enter to continue...
Right Ctrl


Cloud-Service-Provider [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Terminal - dhruv@cloud-ser...
21 Oct, 04:53
Terminal - dhruv@cloud-server: ~/Desktop
File Edit View Terminal Tabs Help
========== SYSTEM LOG ==========
2026-10-21 03:48:34 | Client registered: client01
2026-10-21 03:49:11 | Backup created for client01
2026-10-21 04:02:27 | Backup created for client01
2026-10-21 04:03:40 | Backup created for client01
2026-10-21 04:39:46 | Backup created for client01
Press Enter to continue...
Right Ctrl